GAMIFICATION:
Separating Fact From Fiction

BY KARL KAPP

Four myths have developed about gamification as a learning delivery method. Understanding the truth will determine how, or if, companies should integrate gamification into strategic learning plans.

There once was a myth that the world was flat, and it became a belief for many individuals. However, through research, observation and discovery, the scientific community proved the world is spherical. The focus for learning leaders is similar in that they too must be able to separate fact from myth to determine which tools are effective and which are just trendy.

Consider the concept of gamification. The idea has been held up as either the promise of gaining and forever holding learners' attention, or as a manipulative tool to trick learners into doing something they don't really want to do. The truth is not so dramatic. Gamification is a tool in learning professionals' tool kit with advantages and disadvantages in different situations and environments.

Simply, gamification is an emergent approach to instruction. It facilitates learning and encourages motivation using game elements, mechanics and game-based thinking. Thanks to the hype and misinformation swirling around this learning delivery method, four popular myths have developed. Understanding these can help to avoid pitfalls when integrating gamification into strategic learning plans.

ON THE WEB

Engagement and culture change are two areas where tech-based learning falls short, says Children's Medical Center of Dallas' Justin Lombardo: **CLOmedia.com/materials/view/255.**

TOP SCORES
THIS WEEK
1. JACOB 63730
2. EMMA 62810
3. MICHAEL 60290
4. LIAM 59830
5. SOPHIA 59210

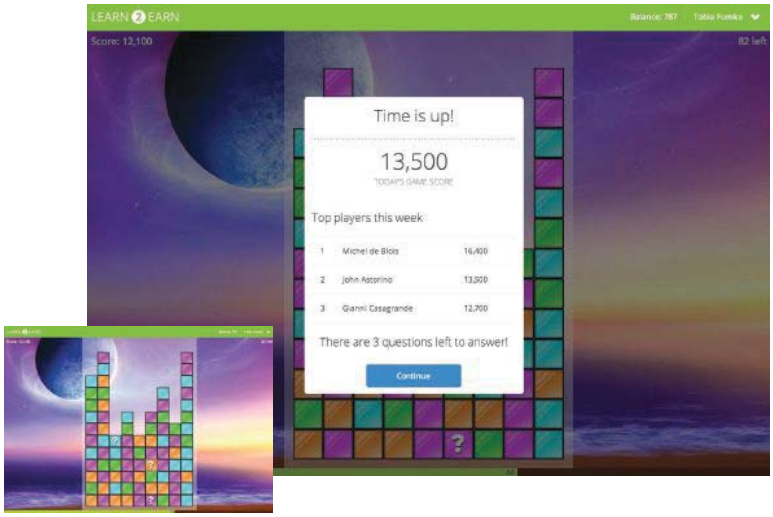
Myth One: Gamification and Games Are the Same

Game-based learning uses an actual game to teach knowledge and skills. A learning game is a self-contained unit with a definitive start, game play and ending. Learners know they are engaged in a game activity, and at the end there is a "win state." Games can deliver different types of learning content in different settings.

Gamification, on the other hand, only uses a few game elements. Learners don't play an entire game from start to finish; they participate in activities that include video or mobile game elements such as earning points, overcoming a challenge or receiving badges for accomplishing tasks.

Game-based learning is often used as a one-time instructional event to provide formal learning either online or within a classroom. Learning games are best suited to teach tradeoffs, resource allocation and decision-making.

Gamification does not typically occur in a classroom; it is often delivered to a learner's computer, tablet or smartphone in two- to five-minute increments. It is a formal structure, but learners can engage with the content when and wherever they happen to be. The content is usually distributed over time, and is not meant to be learned in one setting. Content for gamification varies but is often focused on safety policies, product specifications, customer service, onboarding new employees and other information employees need to be reminded of regularly.



"Block Rush," a short arcade-style game, starts with a screen full of multicolored blocks. The goal is to clear the screen. Learning questions pop up when "?" blocks are selected. Well-designed gamification provides players with tools to monitor their own progress and their progress against others. PHOTOS COURTESY OF AXONIFY

Myth Two: Gamification Alienates Older Learners

It is a myth that people of a certain age do not like games and, therefore, will not like gamification. The flip side is that anyone under age 30 likes games and, therefore, will love all gamification efforts. Neither is true. Some employees don't like online learning, classroom instruction or any kind of training. There is no single training approach that every employee is going to embrace.

> Gamification and games are not the same thing.

This is part of a larger myth that older people don't play video games. According to the Entertainment Software Association's 2013 report "Gamers Over 50 Study: You're Never Too Old to Play," 48 percent of adults age 50 and older say they play video games. Some 80 percent of those play weekly, while 45 percent play daily. The report indicated that gamers who are 50 years old preferred games that mimicked traditional forms of play such as card or tile games (56 percent), puzzle/logic games (52 percent), and trivia, word and board games (27 percent).

Companies can use gamification with employees of all ages on a range of topics including retail and health care sales, and learning about insurance products. For example, Pep Boys, an automotive retail chain, had a 95 percent employee voluntary participation rate in its gamification efforts for employees of all ages.

The average age of employees within a workforce is not an accurate barometer of whether an organization should employ gamification in its learning strategy. A better barometer is to look at what type of content needs to be learned and how often it needs to be reinforced and practiced to have the desired bottom-line impact.

Gamification is helpful for scenarios where leaders have to continually update workforce knowledge and it does not need to occur in a classroom. For instance, a mobile sales force selling technical products such as cellphone services and subscriptions would be an excellent target audience for gamification delivered via mobile device.

READER REACTION

How can learning leaders separate fact from fiction and practically incorporate gaming into their learning strategy?

@naokel79:
I'm using branching scenarios, social learning and collaboration, as well as interactive video tools to teach soft skills.

@metalearner:
Model the system and use simulations to teach decisions and problem resolution.

Mark Goebel:
A strategy must be measurable. The company scorecard is the gamified view of the work of the company. Tracking and racing toward the measurable goals is the essence of gamification!

Jean-Marc Guillemette:
Gamification is the latest example of how we tend to jump on bandwagons or get excited about new tools and trends. Seeing through trends requires keeping learning aligned with corporate objectives. The better organizations define performance improvement in terms of achieving key objectives, the easier it becomes to decide if and how to use games — or anything else — for learning.

Srinivasan Venkatarajan:
Gamification is the buzzword in every organization, but when it comes to gamification in the context of corporate learning, many organization do not know where to start, so they just start implementing something hoping to get some metrics to prove that it worked. If one understands the difference between structural gamification, content gamification and game-based learning, they will be able to make right choice of what, when and how to gamify.

Brad Davis:
There are an amazing number of game mechanics that can be adapted to any theme to add depth while not adding layers of complexity or triviality. Gaming has the power to stick things in your memory in a way few other experiences (short of the real thing) can.

Lester Stephenson:
Gamification is yet another expensive flavor of the month. Save your money. Resist. Resist. Resist.

Richard Butler:
The first thing leaders should do is understand what gamification is and is not, so they can distinguish it from related terms, such as serious games and game-based learning. The key point behind gamification is engagement, so leaders will need to figure out what ideas and incentives capture their learners' attention.

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Employees who have been on the job for several years also could benefit. Often organizations have a well-designed onboarding program where new employees receive a great deal of initial training, but in the subsequent years, companies don't offer as much. Rather than drag them back into the classroom, gamification techniques can be used to refresh their knowledge.

Gamification can also help new employees get to know one another through activities that foster collaboration and cooperation, and learn about the company's policies, vision, mission and products. Software company SAP used gamification to keep recruited students in India interested in the onboarding process by inviting new hires to participate and win medals by answering trivia questions about the company.

Myth Three: There Is No Science Behind Gamification

The most effective gamification platforms use two learning practices — retrieval practice and spaced retrieval. Combined, these techniques provide a strong foundation to increase learning and retention.

Retrieval practice requires learners to recall information rather than re-read or re-listen to the material. Basically, it tests the learner, not for a grade or to evaluate, but to help improve content recall and retention. Research dating back as far as 1907 supports use of retrieval practice as an instructional method. Georgia Southern University professor John Dobson conducted a study in 2013 that found that using a series of very brief retrieval quizzes enhanced retention of previously tested material as much as 40 percent.

Spaced retrieval provides learners with content spaced over time rather than all at once. Researchers Shana Carpenter, Nicholas Cepeda, Doug Rohrer, Sean Kang and Harold Pashler summarized dozens of studies on the topic in their 2012 article "Using Spacing to Enhance Diverse Forms of Learning: Review of Recent Research and Implications for Instruction." The researchers conclude: "The literature reviewed here suggests that in order to promote long-term retention of knowledge, students should receive spaced re-exposure to previously learned information … [especially] if the goal is long-term retention."

When learners receive content on a daily or weekly basis and are quizzed on that content with some additional game elements, the effect is long-term retention and knowledge application. When considering gamifi-

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cation as an organizational learning approach, ensure the sound research-based methods of spaced retrieval and retrieval practice are an integral part of the solution.

Myth Four: Gamification Is About Points, Badges and Leaderboards

The least exciting element of any game is the points, badges or leaderboards. People don't play a game just for points, they play for mastery, to overcome challenges and to socialize with others. The most effective gamification efforts include more than points and badges — they contain elements of story, challenge and continual feedback as well as a high level of interactivity. These are the most engaging elements in games, and they can have a big effect on the organization.

Mike Keeler, vice president of operations at Capital BlueCross, said a well-designed gamification effort can help motivate employees. "They want to know [their scores] versus us wanting to know it. That's a huge change in the learning environment."

Incorporating Gamification Into Learning Strategy

When considering whether to integrate gamification into learning and development strategy, ensure that efforts are not simply a "bolt on" of meaningless, superficial game elements. Employees need to have a structure and a framework to participate in the gamification effort. Clearly describe the challenge before them, provide transparency into how they can be successful and provide an explanation of gamification. These steps can lead to better learning, retention and ultimately increased bottom-line results.

Debunking the common myths of gamification is a good start to intelligently incorporate this approach into a learning strategy. Gamification is best used for subjects that require reinforcement over time and that should be front of mind for employees. Its implementation should be based on the concepts of spaced retrieval and retrieval practice to ensure a solid scientific base. And the focus should be on the learning and expected outcome of the instructional event.

Gamification can provide an edge in learning delivery when it is designed, developed and deployed properly. The effort should not focus solely on points, badges and leaderboards. Learning leaders must understand desired learning outcomes, and structure gamification efforts as an integral part of a larger strategy. CLO

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A Bumpy Road on the Mountain

Iron Mountain did encounter several challenges during the implementation of its Sentinel program. Henry's team conducted three rounds of WebEx conference sessions with regional and market leaders to sustain excitement during the program's initial launch. They received feedback from some of the coaches saying their immediate supervisors weren't providing them adequate time to perform both their coaching duties and their regular responsibilities.

To address this, the team added a fourth round of WebEx sessions for immediate supervisors, discussing how to schedule the coaches to allow them enough time to serve customers while training new hires and addressing supervisors' questions and concerns. "By adding this step, we were able to better prepare the supervisors for the support they needed to provide the coaches when they returned from class," Henry said.

After rollout, the learning team also learned it would need to move quicker to implement the program for other service lines, such as the records centers and document shredding plants. The team adjusted its design and delivery schedule and added resources for coach certification sessions, cutting development time by 50 percent.

The team developed the Sentinel program for the records centers in third quarter 2012 and completed rollout by second quarter 2013. For the shredding plants, program development began second quarter 2013 and rollout was completed in early December. Iron Mountain rolled out the program to its data management service line in February.

At press time, Iron Mountain had deployed nearly 800 total computer terminals in various North American operations locations for the Sentinel program, 117 e-learning modules, 140 on-the-job training modules and had nearly 500 certified coaches.

The company is also starting a rollout for its South American, European, Asian and Australian markets. "We're in this for the long haul and committed to providing the highest levels of service for our customers," Tomovcsik said.

Henry said for her the greatest satisfaction is that the company is investing "in the folks who deliver our service at the front-line levels, the guys and gals who are doing the work every day and are often the face of Iron Mountain for our customers. A lot of companies start peer coaching at salaried levels, but we're investing in front-line hourly employees, and that's something to be really proud about." CLO

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